

Cloud Programming Simplified: A Berkeley View on Serverless Computing

UC Berkeley

A solid blue wave-like shape at the bottom of the slide, starting from the left edge and curving upwards towards the right.

What are two problems with cloud computing

- 1. The appearance of infinite computing resources on demand.
- 2. The elimination of an up-front commitment by cloud users.
- 3. The ability to pay for use of computing resources on a short-term basis as needed.
- 4. Economies of scale that significantly reduced cost due to many, very large data centers.
- 5. Simplifying operation and increasing utilization via resource virtualization.
- 6. Higher hardware utilization by multiplexing workloads from different organizations.

List of low-level virtual machine management responsibilities

- 1. Redundancy for availability, so that a single machine failure doesn't take down the service.
- 2. Geographic distribution of redundant copies to preserve the service in case of disaster.
- 3. Load balancing and request routing to efficiently utilize resources.
- 4. Autoscaling in response to changes in load to scale up or down the system.
- 5. Monitoring to make sure the service is still running well.
- 6. Logging to record messages needed for debugging or performance tuning.
- 7. System upgrades, including security patching.
- 8. Migration to new instances as they become available.

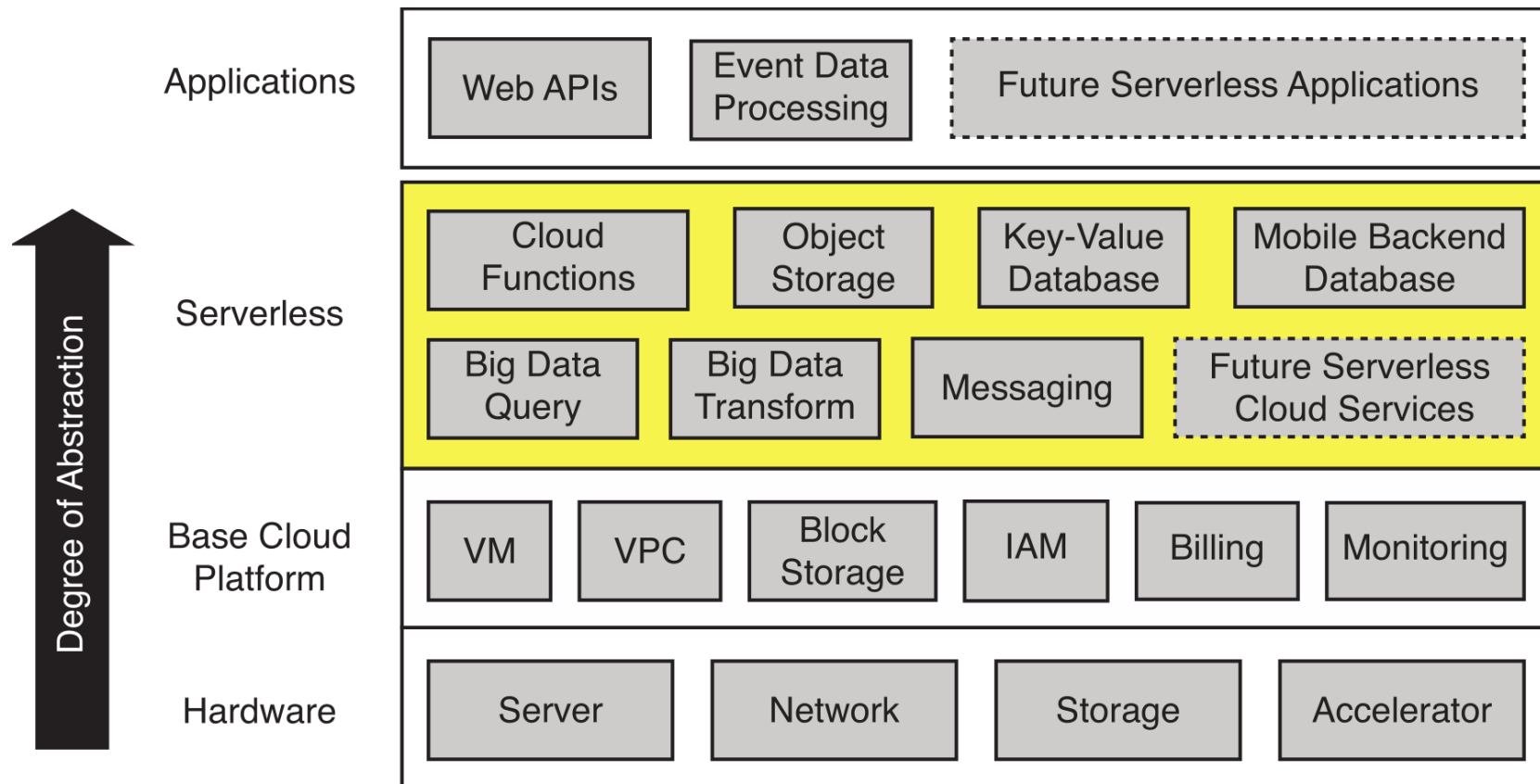
What is Serverless Computing ?

- Serverless computing = FaaS + BaaS

What is Serverless Computing ?

- In any serverless platform, the user just writes a cloud function in a high-level language, picks the event that should trigger the running of the function—such as loading an image into cloud storage or adding an image thumbnail to a database table

What is Serverless Computing ?



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- Decoupled computation and storage
- Executing code without managing resource allocation
- Paying in proportion to resources used instead of for resources allocated

How serverless computing differs from its predecessors (e.g PaaS)

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- Better autoscaling
- strong isolation
- platform flexibility
- service ecosystem support.

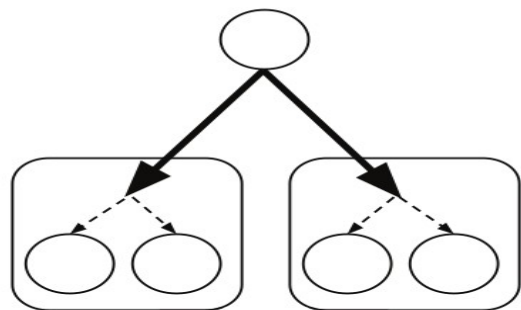
Limitations of Today's Serverless Computing Platforms

<i>Percent</i>	<i>Use Case</i>
32%	Web and API serving
21%	Data Processing, e.g., batch ETL (database Extract, Transform, and Load)
17%	Integrating 3rd Party Services
16%	Internal tooling
8%	Chat bots e.g., Alexa Skills (SDK for Alexa AI Assistant)
6%	Internet of Things

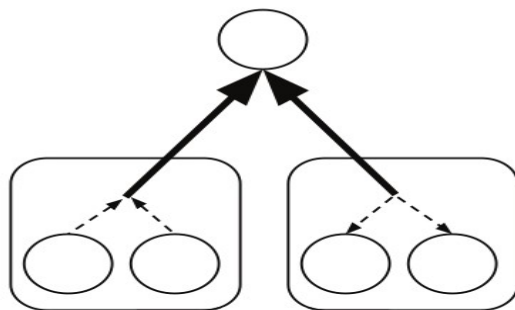
Limitations of Today's Serverless Computing Platforms

- Inadequate storage for fine-grained operations
- Lack of fine-grained coordination
- Poor performance for standard communication patterns
- Predictable Performance

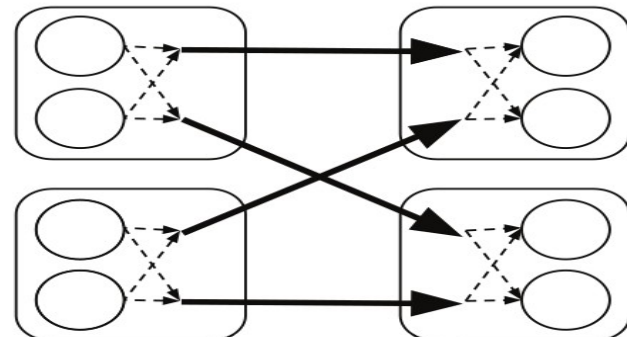
- **Poor performance for standard communication patterns**



Broadcast

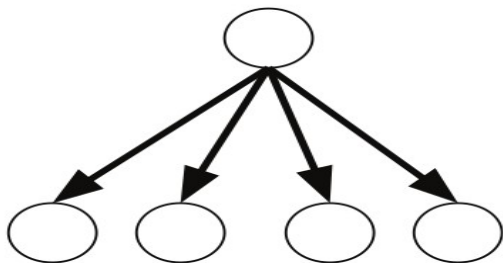


Aggregation

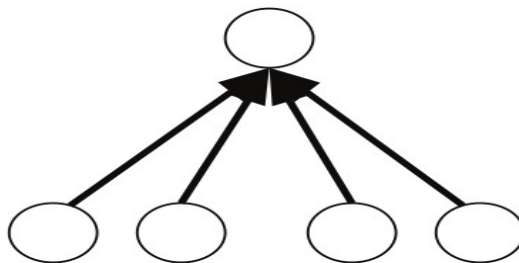


Shuffle

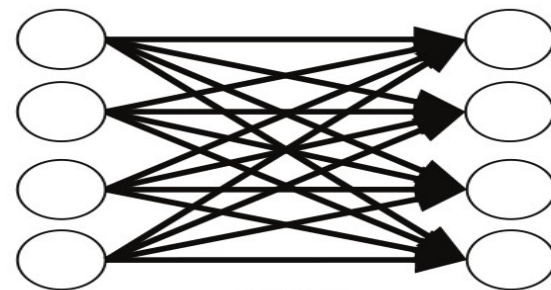
(a) VM-based communication patterns.



Broadcast



Aggregation



Shuffle

Abstraction challenges

Abstraction challenges

- Resource requirements

Abstraction challenges

- Resource requirements
- Data dependencies

System challenges

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- High-performance, affordable, transparently provisioned storage
 - Ephemeral Storage
 - Durable Storage

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- High-performance, affordable, transparently provisioned storage
 - Ephemeral Storage
 - Durable Storage
- Coordination/signaling service
- Minimize startup time

Networking challenges

Networking challenges

- cloud functions can impose significant overhead on popular communication primitives such as broadcast, aggregation, and shuffle

Security challenges

Security challenges

- Scheduling randomization and physical isolation
- Fine-grained security contexts
- Oblivious serverless computing

Computer architecture challenges

Computer architecture challenges

- Hardware Heterogeneity, Pricing, and Ease of Management

Fallacies and Pitfalls

- **Fallacy** Since an AWS Lambda cloud function instance with equivalent memory capacity of an on-demand AWS t3.nano instance (0.5 GiB) costs 7.5x as much per minute, serverless cloud computing is more expensive than serverful cloud computing
- **Pitfall** Serverless computing can have unpredictable costs
- **Fallacy** Since serverless computing programming is in high-level languages like Python, it's easy to port applications between serverless computing providers

Fallacies and Pitfalls

- **Pitfall** Vendor lock-in may be stronger with serverless computing than for serverful computing
- **Fallacy** Cloud functions cannot handle very low latency applications needing predictable performance
- **Pitfall** Few so called “elastic” services match the real flexibility demands of serverless computing

End